

Cerebral Amyloid Angiopathy
Kirthivasan Ramanathan
r.kirthivasan@nhs.net
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[The Link to the PDF version](#)

The Link to the HTML5 version: https://mj-ramirez.github.io/Cerebral_Angiopathy_talk/

Technical Support & Fact-Checking:
Manfred Junemann Ramirez

CASE PRESENTATIONS

- 3 tales of Thrombosis, haemorrhage and inflammation
- One unifying diagnosis

CASE NUMBER 1: A 65 YEARS OLD FEMALE

PRESENTATION

- Sept 2013 RIGHT hemispheric event
 - Intermittent confusion, subtle left facial weakness
 - CT: extensive hypodensity right temporoparietal lobe
 - MRI: DWI hyperintensity/T2 and FLAIR hyperintensities right temporal lobe – no enhancement
 - Surveillance MRI revealed complete resolution
- April 2020 LEFT hemispheric event
 - Right arm numbness and aphasia for an hour followed by confusion
 - CT head showed oedema in left temporoparietal area
 - MRI revealed now LEFT sided changes
 - MRI repeated 2 months later - resolution

IMAGING



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CASE NUMBER 2: A 67 YEARS OLD MALE

- PMH: on anticoagulant for Anti Phospholipid Antibody syndrome

PRESENTATION

- Jan 2019 LEFT hemispheric event
 - Sudden onset of pins and needles right hand
 - Imaging revealed left focal SAH
- July 2024 recurrent LEFT hemispheric event
 - 2 x episodes of stereotyped events one hour apart: right hand tingling and slurred speech
 - Whilst an inpatient had recurrent episode of right facial numbness, tingling of right hand and slurred speech
 - Resolved and home on continued treatment
- Jan 2025 recurrent LEFT hemispheric event
 - Several episodes of recurrent stereotyped episodes of right facial numbness, tingling in the right hand and slurred speech

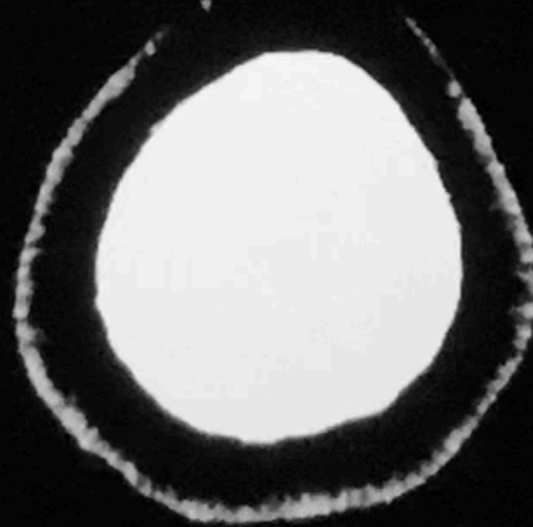
IMAGING



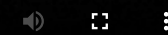
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CT Head January 2019

Sequence 1 of 4



▶ 0:00 / 0:38



CASE NUMBER 3: A 65 YEARS OLD FEMALE

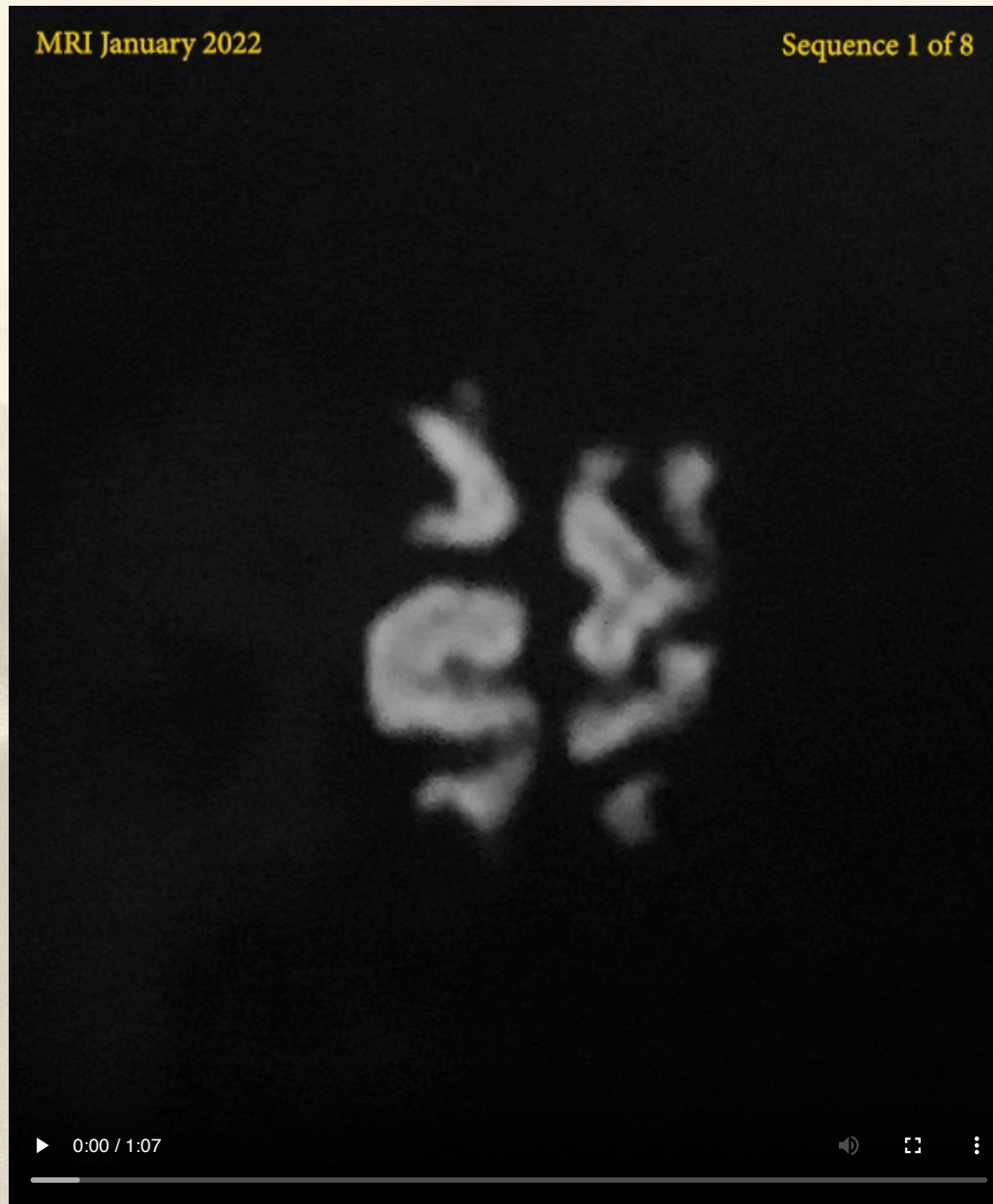
PRESENTATION

- July 2022 RIGHT focal cortical infarct
 - Two episodes of Left facial droop, left sided paresthesia and dysarthria
- May 2023 LEFT focal cortical infarct
 - Transient right hemifield visual loss
- September 2023 recurrent RIGHT focal cortical infarct
 - Left lower quadrantanopia, left arm drift
- January 2025
 - New onset atrial fibrillation
 - Tingling of the left hand
 - MRI revealed moderate white matter hyperintensities
 - Incidental cortical microbleeds in Lobar distribution L>R

IMAGING



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CEREBRAL AMYLOID ANGIOPATHY

- Cerebral amyloid angiopathy (CAA) is a chronic, degenerative cerebrovascular disease caused by deposition of amyloid protein in cerebral vessel walls.

CRITERIA FOR DIAGNOSIS

Charidimou, Boulouis & Frosch et al. (2022) The Boston Criteria Version 2.0 for Cerebral Amyloid Angiopathy: A Multicentre, Retrospective, MRI-neuropathology Diagnostic Accuracy Study, The Lancet. Neurology. [PubMed](#)

CLINICAL PRESENTATION

- Progressive cognitive decline
- Any stroke presentation including Hemiplegia
- Transient focal neurological episodes

TRANSIENT FOCAL NEUROLOGICAL EPISODES (TFNES)

- Brief disturbances of motor, somatosensory, visual or language functions
- Frequently recurrent stereotyped episodes of focal disturbances
- Often presenting of transient paresthesia of the mouth or hand
- Frequently combining sensory and motor symptoms
- Very close resemblance to TIAs
- They pose a high risk for ICH and correlate with focal SAHs and cSS

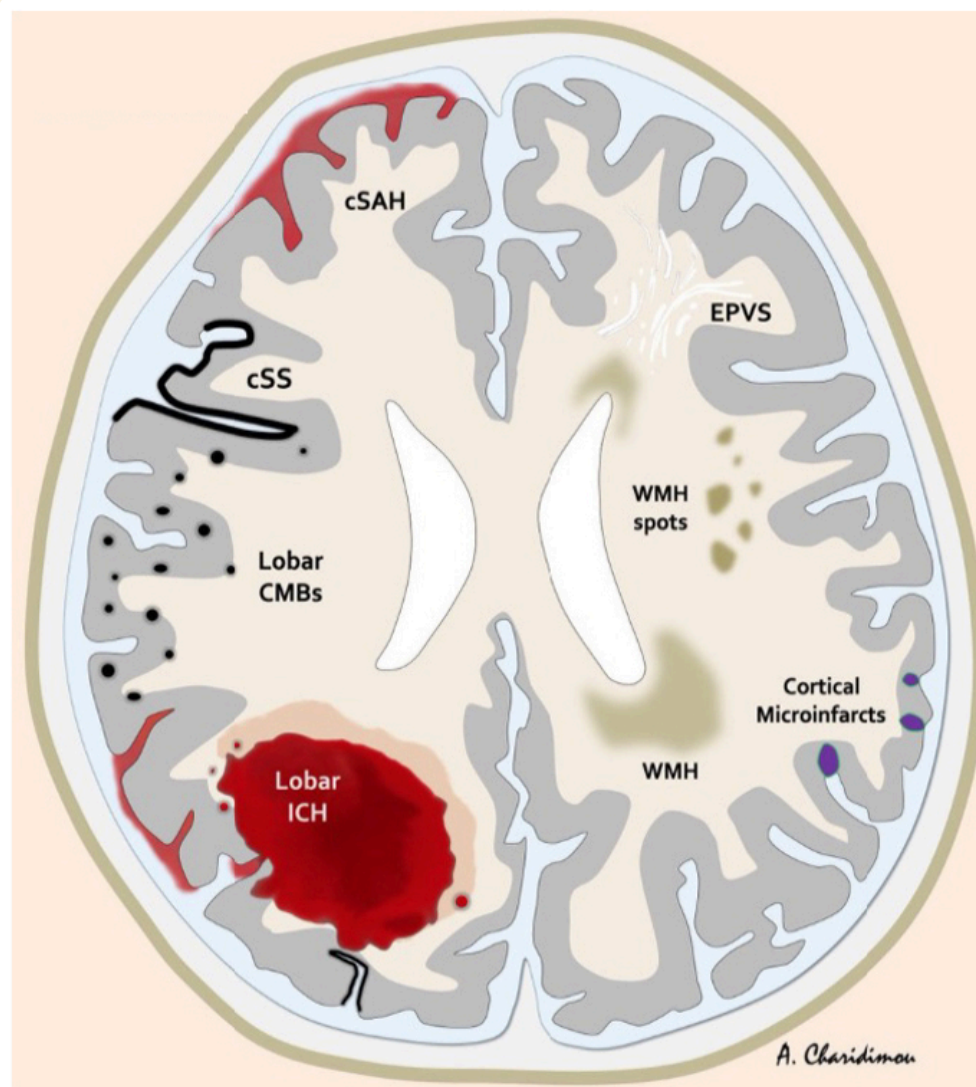
IMAGING

- CT Head
- MRI
- MR Fast Field Echo magnetic resonance imaging (FFE)
- MR Gradient Echo Sequence magnetic resonance imaging (GRE)
- MR Susceptibility-weighted magnetic resonance imaging (SWI)
- MR Diffusion-weighted magnetic resonance imaging (DWI)
- MR T2-weighted-Fluid-Attenuated Inversion Recovery (T2-FLAIR)

IMAGING FEATURES OF CEREBRAL AMYLOID ANGIOPATHY

- Cortico-subcortical microhaemorrhages or petechial haemorrhages (CMB)
- Focal cortical subarachnoid haemorrhage (SAH)
- Disseminated cortical superficial siderosis (cSS)
- Lobar ICH with micro bleeds
- Large and confluent areas of white matter hyperintensities in T2/FLAIR-weighted images and scattered microbleeds on GRE/SWI sequences
- (Recurrent) lobar ICH - with finger like projections
- White matter hyper intensities (Cortical, Subcortical, Focal)
- Focal Cortical Microinfarcts

RADIOLOGICAL FINDINGS IN CAA



Haemorrhagic:

- Convexity subarachnoid hemorrhage (cSAH)
- Lobar Intracranial Haemorrhage (ICH)
- Cerebral micro bleeds (CMB)
- Cortical superficial siderosis (cSS)

Non haemorrhagic:

- Enlarged Perivascular Spaces (EPVS)
- Cortical microinfarcts
- White matter hyperintensities (WMH)

PATHOLOGY

- Amyloid- β deposition in cortical/leptomeningeal vessels (intra operative or post mortem)
- CSF analysis combined findings of:
 - 50% reduction in levels of amyloid β -42 and β -40 protein
 - Mildly increased total τ (tau)-protein levels

GENETIC

- Apolipoprotein E (APOE) ϵ 4 and ϵ 2 allele (risk factor)
- Family history of CAA or hereditary forms (e.g. Amyloid Precursor Protein mutations)

SUPPORTING FEATURES

Not required but suggestive

- Neuroimaging:
 - Acute convexity subarachnoid heamorrhage (non-traumatic)
 - Severe white matter hyperintensities (posterior-predominant)
 - Enlarged perivascular spaces (centrum semiovale >20 count)
- Genetic:
 - APOE ϵ 4 allele (risk factor)
 - Family history of CAA or hereditary forms (e.g. APP mutations)

POSSIBLE CEREBRAL AMYLOID ANGIOPATHY

- Age ≥ 50 years
- MRI evidence of one strictly lobar heamorrhage (ICH, CMB, or cSS)
- No alternative cause identified

PROBABLE CEREBRAL AMYLOID ANGIOPATHY

- Age ≥ 50 years
- MRI evidence of ≥ 1 of the following
- ≥ 2 strictly lobar heamorrhages: spontaneous ICH, cerebral microbleeds (CMBs), or cortical superficial siderosis (cSS)
- One lobar heamorrhage + focal/disseminated cSS (≥ 3 sulci)
- Exclusion of other causes (e.g. hypertension, anticoagulation, trauma, heamorrhagic tumor)

DEFINITE CEREBRAL AMYLOID ANGIOPATHY

- Requires postmortem confirmation
- Pathological evidence of amyloid- β deposition in cortical/leptomeningeal vessels
- Documentation of lobar haemorrhage: macro/microbleed, cortical superficial siderosis (cSS), or intracerebral haemorrhage (ICH)

CEREBRAL AMYLOID ANGIOPATHY RELATED INFLAMMATION

- CAA-RI is often diagnosed in the context of the modified Boston criteria for CAA, which include:
 - Probable or definite CAA based on imaging or pathology.
 - Evidence of inflammation (e.g. white matter changes, edema) not explained by other causes.
- The diagnosis of CAA-RI is based on a combination of clinical, imaging, and sometimes histopathological findings.

CORTICAL MICROBLEEDS

Cerebral small vessel diseases	Other brain disease	Systemic diseases
• Arteriolosclerosis • Sporadic cerebral amyloid angiopathy (CAA) • Inflammatory CAA • Iatrogenic CAA • Alzheimer's disease (with CAA) • Genetic small vessel diseases • CADASIL • CARASIL • CARASIL • COL41A • Hereditary CAA • Fabry disease	• Moya–Moya syndrome • CNS vasculitis • Radiation vasculopathy • Reversible Cerebral Vasoconstriction Syndrome • (Multiple sclerosis)	• Infective endocarditis • Cardiac valve replacement / cardiopulmonary bypass • Antiphospholipid syndrome • Sneddon syndrome • Sickle disease • Thrombotic thrombocytopenic purpura • Hemophilias • Critical care encephalopathy • COVID-19 • ECMO • Chronic renal impairment

SUPERFICIAL SIDEROSIS

CLASSIC SUPERFICIAL SIDEROSIS	CORTICAL SUPERFICIAL SIDEROSIS
<u>Neurosurgery</u>	<u>Amyloid angiopathy</u>
Meningocele	<u>Cranioencephalic traumas</u>
Trauma, <u>posttraumatic cyst</u>	<u>Reversible vasoconstriction syndrome</u>
Vasculitis	<u>Ruptured aneurysm</u>
<u>Bleeding neoplasm</u>	<u>Interhemispheric arteriovenous fistulas</u>
Intracranial subdural hematoma	<u>CNS angiitis</u>
<u>Unruptured aneurysm</u>	<u>Infectious endocarditis</u>
<u>Arteriovenous malformation</u>	<u>Postrevascularization hyperperfusion syndrome</u>
<u>Cavernous malformation</u>	<u>Dural AV fistulas</u>
<u>Neonatal intraventricular hemorrhage</u>	<u>Cortical vein thrombosis</u>
<u>Oral anticoagulant therapy</u>	

MANAGEMENT AND TREATMENT OPTIONS

THE ANTITHROMBOTIC SWOT ANALYSIS

Strength

Weakness

Thrombotic indications

Hemorrhagic risk profile

Opportunity

Threat

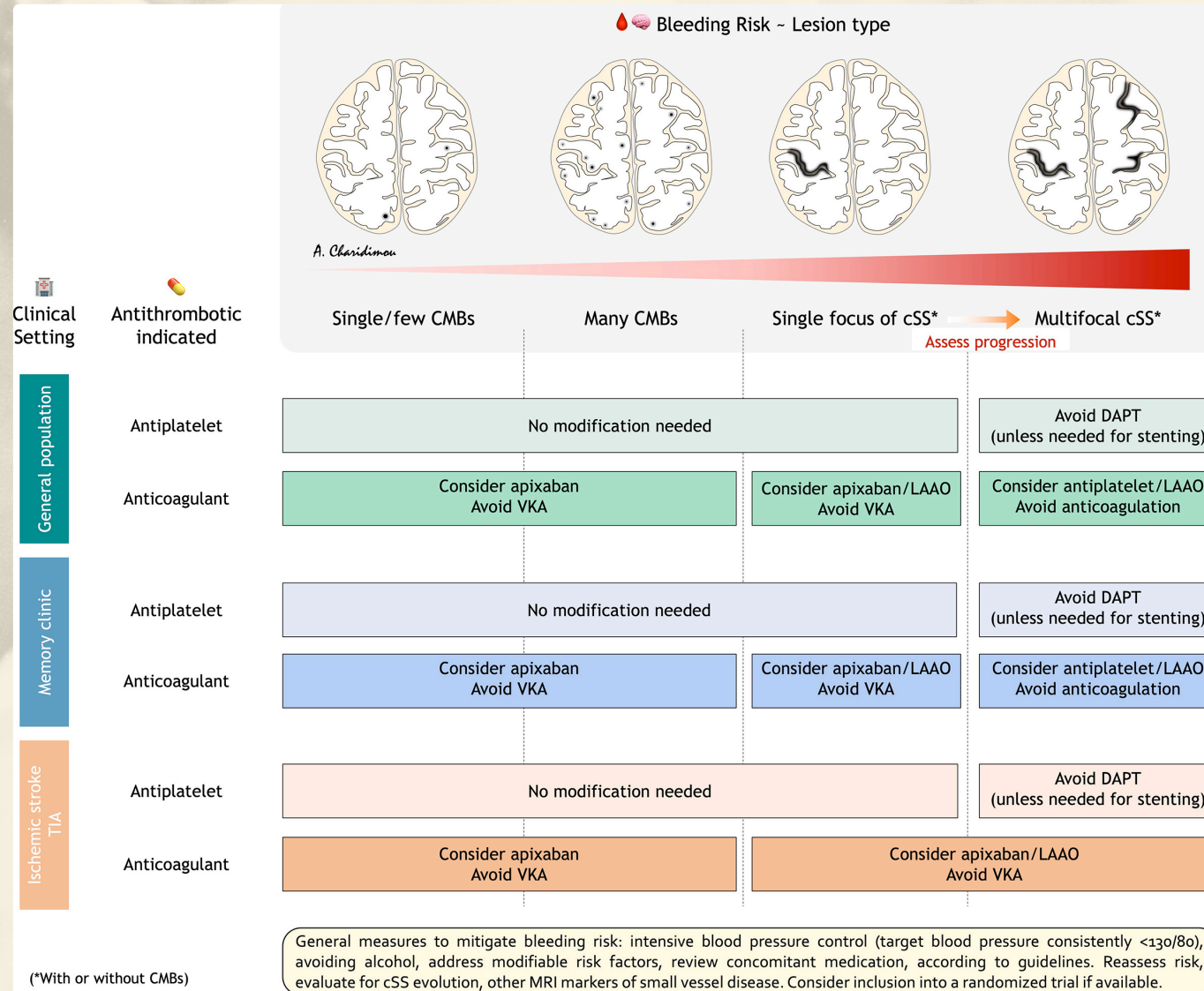
Optimisation of thrombotic
risk

Adverse outcomes

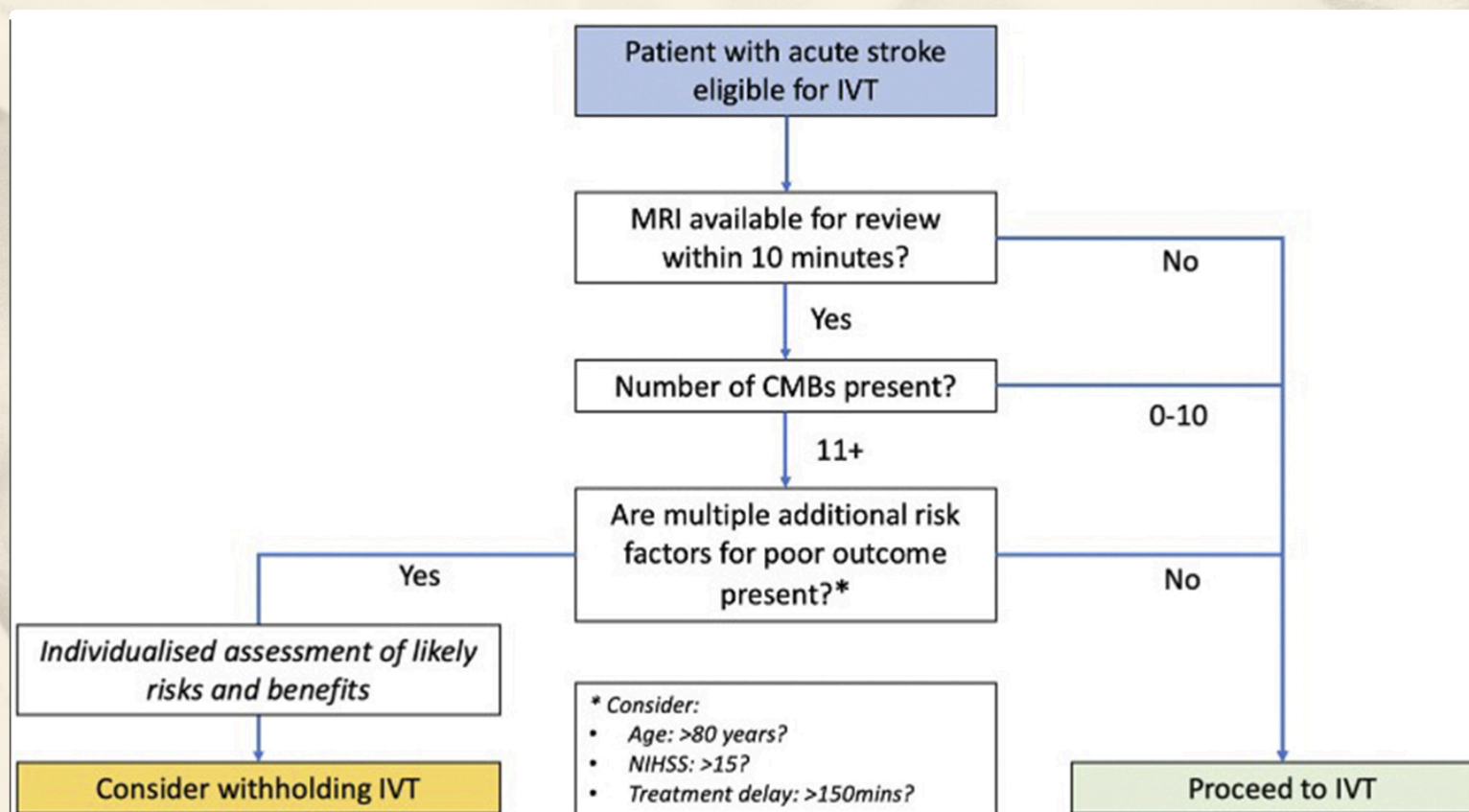
RISK OF INTRACRANIAL BLEEDING

Figure 1: Managing antithrombotics and anticoagulants in patients with asymptomatic microbleeds (CMBs) and Cortical superficial siderosis (cSS).

Reference: Charidimou & Smith (2024) Cardiovascular Management in Asymptomatic (Silent) Cerebral Microbleeds and Suspected Cerebral Amyloid Angiopathy, Stroke.



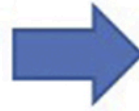
CMB INTRACRANIAL BLEEDING RISK I



Best, Jesuthasan & Werring (2023) Cerebral Small Vessel Disease and Intracranial Bleeding Risk: Prognostic and Practical Significance, International Journal of Stroke. [PubMed](#)

CMB INTRACRANIAL BLEEDING RISK II

Variable		Score
Age	<70	0
	70-79	3
	80+	4
CMB burden	0	0
	1-4	3
	5-10	5
	11-19	6
	20+	9
2D T2* GRE sequence used		2
Previous intracranial haemorrhage		5
East Asian population		2
Ischaemic stroke before presenting stroke/TIA		1
Antiplatelet or VKA treatment		1
TOTAL		/24



Total Score	Risk Category	Predicted five-year risk	Observed risk in internal validation cohort (95% CI)
0 -4	Lower	1.0%	0.9 (0.6 - 1.3)
5 - 7	Intermediate	2.3%	2.5 (1.8 - 3.5)
8+	Highest	6.7%	6.5 (4.8 - 8.7)

Best, Jesuthasan & Werring (2023) Cerebral Small Vessel Disease and Intracranial Bleeding Risk: Prognostic and Practical Significance, International Journal of Stroke. [PubMed](#)

HEAMORRHAGE AFTER THROMBOLYSIS (HAT) SCORING SYSTEM

3 criteria, 4 points max

- The Points:
 - Prior intra cranial heamorrhage (ICH) = 2 points
 - cortical Superficial Siderosis (cSS) = 1 point
 - APOE $\epsilon 4/\epsilon 2$ = 1 point
- The Interpretation:
 - 0 points = 3.7 risk of bleed
 - 3 points = 25% risk

Reference: Lou, Safdar & Mehdiratta et al. (2008) The HAT Score: A Simple Grading Scale for Predicting Heamorrhage after Thrombolysis, Neurology. [PubMed](#)

ANTITHROMBOTIC THERAPY AFTER INTRACEREBRAL HAEMORRHAGE

- RESTART trial: Anticoagulation [PubMed](#)
 - The notion that anticoagulation should be permanently avoided following ICH has been challenged and a more nuanced approach is suggested
 - One such nuance however is that certain phenotypes of CAA confer the highest rebleed risk
- Antiplatelets (More controversial)
 - RESTART AP/SoSTART – small and underpowered but initial results are appearing no increase in risk. [ASPIRING](#) is recruiting and we are participating
 - Metaanalysis suggests modest increase in ICH with antiplatelets with an RR 1.3-1.5 but is it outweighed by prevention of thrombotic events? JURY IS OUT

NOVEL TREATMENT MODALITIES

- Monoclonal Antibodies Targeting Amyloid- β
- β -Secretase (BACE) Inhibitors
- γ -Secretase Modulator
- Amyloid Aggregation Inhibitors
- Active Immunotherapy

PROGNOSIS AND OUTCOMES

- There is no cure
- Managing hypertension and lifestyle modifications can slow progression
- Avoiding anticoagulant to reduce heamorrhage risk

QUIZ

WHAT ARE TYPICAL IMAGING FEATURES OF CAA (TRUE OR FALSE?)

- cortico-subcortical microhaemorrhages or petechial haemorrhages
- Large and confluent areas of white matter hyperintensities in T2/FLAIR-weighted images and scattered microbleeds on gradient-echo sequences
- (Recurrent) lobar ICH - with finger like projections
- Focal cortical SAH
- Subdural haemorrhages
- Basal ganglia haemorrhages

IMAGING FEATURES ASSOCIATED WITH HIGHER RISK OF ICH (TRUE OR FALSE?)

- Severe white matter hyper intensities
- Disseminated cortical superficial siderosis
- Micro bleeds
- Cortical atrophy

TRANSIENT FOCAL NEUROLOGICAL EPISODES (TRUE OR FALSE?)

- Brief disturbances of motor, somatosensory, visual or language functions
- Frequently recurrent stereotyped episodes of focal disturbances
- Often presenting of transient paraesthesia of the mouth or hand
- Very close resemblance to TIAs
- As these are typically thrombotic, they do not pose a ICH risk

REFERENCES

Lou, Safdar & Mehdiratta et al. (2008) The HAT Score: A Simple Grading Scale for Predicting Hemorrhage after Thrombolysis, Neurology. [PubMed](#)

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Chiti & Dobson (2017) Protein Misfolding, Amyloid Formation, and Human Disease: A Summary of Progress Over the Last Decade, Annual Review of Biochemistry. [PubMed](#)

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- Thank You !!!

A background image showing several medical professionals in white scrubs and masks, looking down, likely in a clinical setting. The image is faded and serves as a backdrop for the text.

THANK YOU !!!